Analisi Matematica 2
Foglio di esercizi n. 5

1. Determinare il massimo e il minimo delle seguenti funzioni negli insiemi indicati.

a. $f(x,y,z) = x + z$ in $D = \{(x,y,z) : x^2 + y^2 + z^2 = 2, z = x^2 + y^2\}$

b. $f(x,y,z) = 2x + 2y - z$ in $D = \{(x,y,z) : x^2 + y^2 + z^2 \leq 4\}$

c. $f(x,y,z) = 4x + z^2$ in $D = \{(x,y,z) : x^2 + y^2 \leq z \leq 4\}$

2. Calcolare i seguenti integrali doppi.

a. $\displaystyle\iint_D \frac{e^{x/y}}{y^4}\,dxdy$ con $D = [0,2] \times [1,2]$

b. $\displaystyle\iint_D (1 - |x| - y)\,dxdy$ con $D = \{(x,y) : x \geq -1, y \geq 0, x + y \leq 1\}$

c. $\displaystyle\iint_D \frac{1}{(2y - x + 8)^2}\,dxdy$ con $D = \{(x,y) : y^2 \leq x \leq 2y + 3\}$

d. $\displaystyle\iint_D \frac{x + y}{1 + x^2}\,dxdy$ con $D = \{(x,y) : y \geq |x| - 1, x^2 + y^2 \leq 1\}$

e. $\displaystyle\iint_D y(1 - \sin(x))\,dxdy$ con $D = \{(x,y) : y \geq 0, 1 \leq |x| + y \leq 2\}$

f. $\displaystyle\iint_D \sqrt{1 - x^2 - y^2}\,dxdy$ con $D = \{(x,y) : y \geq x, x^2 + y^2 \leq 1\}$

g. $\displaystyle\iint_D (x - 1)^2\,dxdy$ con $D = \{(x,y) : x^2 + 4y^2 \leq 1\}$

h. $\displaystyle\iint_D (x - y)\log(x + y)\,dxdy$ con $D = \{(x,y) : 0 \leq x - y \leq 1 \leq x + y \leq 3\}$

3. Calcolare il volume dei seguenti insiemi.

a. $D = \left\{(x,y,z) : \frac{|x|}{4} + \frac{|y|}{3} + |z| \leq 1\right\}$

b. $D = \{(x,y,z) : x^2 + y^2 + z^2 \leq 4, z \geq 1\}$

c. $D = \{(x,y,z) : 0 \leq 4 - x^2 - y^2 \leq z \leq 1\}$

d. $D = \{(x,y,z) : 3(x^2 + y^2) \leq z^2, x^2 + y^2 + (z - 2)^2 \leq 4\}$

4. Calcolare i seguenti integrali tripli.

a. $\displaystyle\iiint_D \frac{1}{(x + y + z + 1)^3}\,dxdydz$ con $D = \{(x,y,z) : x \geq 0, y \geq 0, z \geq 0, x + y + z \leq 1\}$

b. $\displaystyle\iiint_D |y - x|e^z\,dxdydz$ con $D = \{(x,y,z) : x,y,z \in [0,1], z \leq x + y \leq 1\}$

c. $\displaystyle\iiint_D ze^{x^2+y^2}\,dxdydz$ con $D = \{(x,y,z) : x^2 + y^2 + z^2 \leq 1, x \geq 0, y \geq 0, z \geq 0\}$

d. $\displaystyle\iiint_D z^2\,dxdydz$ con $D = \{(x,y,z) : x^2 + y^2 \leq 1 - z^2, x^2 + y^2 \leq 2z - z^2)\}$

e. $\displaystyle\iiint_D |x - 1|\,dxdydz$ con $D = \{(x,y,z) : x^2 + y^2 + z^2 \leq 4, x \geq 0, z \geq 0\}$